II. Smooth functions and smooth maps

1 Basic property

Prop II.1.1. $f: M \to N$ is smooth. Then f is continuous.

proof: $\forall x \in M$, $\exists$ smooth charts (U, φ), (V, ψ) s.t. $f(U) \subseteq V$

$\psi \circ f \circ \varphi^{-1}: \varphi(U) \to \psi(V)$ is smooth. Then $f|_U = \psi^{-1} \circ (\psi \circ f \circ \varphi^{-1}) \circ \varphi: U \to V$ continuous

Since f is continuous in $Nbh(x)$ $\forall x \in M \Rightarrow f$ is continuous $\square$

Prop. II.1.2. $f: M \to N$ is smooth iff $\forall p \in M$ $\exists$ (U, φ) containing p. (V, ψ) containing $f(p)$

s.t. $U \cap f^{-1}(V)$ open in M, $\psi \circ f \circ \varphi^{-1}: \varphi(U \cap f^{-1}(V)) \longrightarrow \psi(V)$ is smooth

Pf: $\Leftarrow$ let $U' = U \cap f^{-1}(V)$ then $f(U) \subseteq V$. let $\varphi': U' \to \varphi(U')$ $\varphi' = \varphi|_{U'}$

then (U', φ') is a smooth chart containing , $\psi \circ f \circ \varphi'^{-1} = \psi \circ f \circ \varphi|_{U'}^{-1}: \varphi(U') \to \psi(V)$ smooth

$\Rightarrow$ Since we can find (U, φ), (V, ψ) s.t. $f(U) \subseteq V \Rightarrow U \cap f^{-1}(V) = U$. $\square$

Prop. II.1.3. (Smoothness is local). M, N be smooth manifolds, $f: M \to N$. then

(a) If $\forall p \in M$. $\exists U$ open containing p, s.t. $f|_U$ smooth, then f is smooth

(b) If f is smooth, then $f|_U$ is smooth, $\forall U \subseteq M$ open.

Proof: (a). $\forall x \in M$. $\exists U$ open. s.t. $f|_U$ smooth.

$\exists U'$ containing x. (V', ψ), $f(U' \cap U) \subseteq V'$ $\psi \circ f|_U \circ \varphi^{-1}: \varphi(U') \to \psi(V')$ smooth

That's $f(U' \cap U) \subseteq V'$, $\psi \circ f \circ \varphi^{-1}: \varphi(U' \cap U) \to \psi(V')$ smooth $\Rightarrow f$ is smooth.

(b) $\forall p \in U$. $\exists (U'', \varphi')$ containing p. (V'', ψ') s.t. $f(U'') \subseteq V''$

$\Rightarrow \psi' \circ f \circ \varphi'^{-1}: \varphi'(U') \to \psi'(V')$ is smooth

then $\psi' \circ f|_U \circ \varphi'^{-1}|_{\varphi'(U \cap U')} = (\psi' \circ f \circ \varphi'^{-1}) \circ \varphi' \circ \varphi^{-1}|_{U \cap U'}: \varphi(U \cap U') \to \psi'(V')$ is smooth

$\Rightarrow f|_U$ is smooth $\square$.

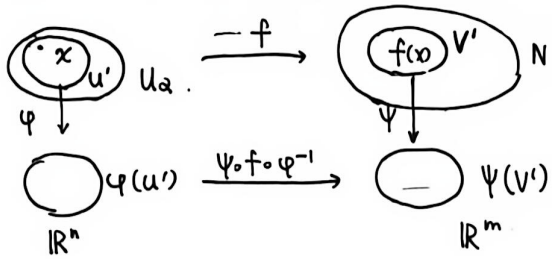
Coro. II.1.4 (Gluing lemma) M, N be smooth, $\{U_\alpha\}_{\alpha \in A}$ be open cover of M, $\forall \alpha \in A$.

give a smooth map $f_\alpha: U_\alpha \to N$ with $f_\alpha|_{U_\alpha \cap U_\beta} = f_\beta|_{U_\alpha \cap U_\beta}$ $\forall \alpha, \beta \in A$

Then $\exists f: M \to N$ smooth. s.t. $f|_{U_\alpha} = f_\alpha$ $\forall \alpha \in A$

proof: Define $f: M \to N$ $x \mapsto f_\alpha(x)$. (By the open over of M).

Then $f|_{U_\alpha}$ is smooth. $\forall \alpha \in A$. $\Rightarrow f$ is smooth.



Prop II.1.5 (1) If $F: M \to N$, $g: N \to P$ smooth then $g \circ f: M \to P$ is smooth

(2) If $M_1, \dots, M_k$, N smooth. Then $F: N \to M_1 \times \dots \times M_k$ is smooth

iff $F_i := \pi_i \circ F: N \to M_i$ is smooth

proof (skipped) □

2 Partitions of Unity and its applications

Def II.2.1. M is a topological space, let $\mathcal{X} = (X_\alpha)_{\alpha \in A}$ be arbitrary open cover of M.

An partition of unity subordinate to $\mathcal{X}$ is a family of continuous functions

$\{\psi_\alpha\}_{\alpha \in A}$ $\psi_\alpha: M \to \mathbb{R}$ s.t.

(i). $0 \le \psi_\alpha \le 1$. (ii) $\operatorname{supp} \psi_\alpha \subseteq X_\alpha$ (iii) $\{\operatorname{Supp} \psi_\alpha\}_{\alpha \in A}$ is locally finite;

(iv) $\sum_{\alpha \in A} \psi_\alpha(x) = 1 \quad \forall x \in M$

Thm II.2.2. M^n is a smooth manifold (with or without boundary), let $\mathcal{X} = (X_\alpha)_{\alpha \in A}$ be arbitrary open cover of M. Then there exists a smooth partition of unity subordinate to $\mathcal{X}$

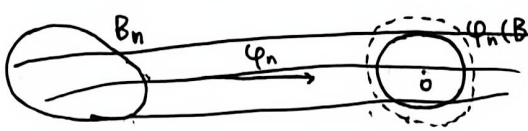
Lemma II.2.3. $B \subseteq M$ is a regular coordinate ball if $\exists$ coordinate ball $B' \supseteq \overline{B}$, $\varphi: B' \to \mathbb{R}^n$

s.t. $\exists r' > r$. $\varphi(B) = B_r(0)$ $\varphi(\overline{B}) = \overline{B}_r(0)$, $\varphi(B') = B_{r'}(0)$.

Every smooth manifold has a countable basis of regular coordinate balls

proof: ~~M^n has a countable basis of paracompact coordinate balls $\{B_n\}_n$ $\{(B_n, \varphi_n)\}_n$~~

~~W.L.O.G. $\varphi_n(B_n)$ has the center of the origin.~~



It's easy to construct such regular balls.

Back to the proof: W.L.O.G. M^n is a smooth manifold without boundary.

$\forall \alpha \in A$. $\exists \mathcal{B}_\alpha$ of regular coordinate balls

$\Rightarrow \mathcal{B} = \bigcup_{\alpha \in A} \mathcal{B}_\alpha$ is a basis of M.

Since M is paracompact, $\exists$ countable, locally finite refinement $\{B_n\}$

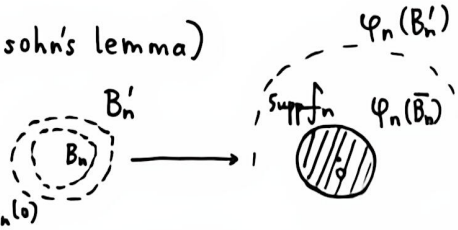
By the local finiteness, $\{\overline{B}_n\}$ is still locally finite

Since $B_n \subseteq X_\alpha$ is regular, there exists $X_\alpha \supseteq B_n' \supseteq \overline{B}_n$, s.t. $\varphi_n: B_n' \to \mathbb{R}^n$

s.t. $\varphi_n(\overline{B}_n) = \overline{B}_{r_n}(0)$, $\varphi_n(B_n') = B_{r_n'}(0)$, $r_n < r_n'$

Define $f_n: M \to \mathbb{R}^{\ge 0}$ by

$$f_n = \begin{cases} H_n \circ \varphi_n & \text{on } B_n' \\ 0 & \text{on } M \setminus \overline{B}_n \end{cases}$$ (By Urysohn's lemma)



where H_n is smooth, $H_n(x) > 0 \quad \forall x \in B_{r_n}(0)$

$H_n(x) = 0 \quad \forall x \in B_{r_n'}(0) \setminus B_{r_n}(0)$

$\operatorname{supp} f = \overline{B}$

Now we can define $f: M \to \mathbb{R}$ by $f(x) = \sum_{n=1}^{\infty} f_n(x)$

. Because of local finitnoss of $\{\overline{B_n}\}$, $\forall x \in M$, it's just a finite sum, thus smooth

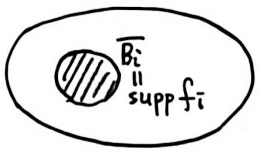
Then $f(x) > 0 \quad \forall x \in M$ let $g_i : M \to \mathbb{R} \quad x \mapsto \frac{f_i(x)}{f(x)}$ are smooth.

$\Rightarrow 0 \le g_i \le 1$, $\sum_{i=1}^{\infty} g_i \equiv 1$.

By axiom of choice, choose $a(i) \in A$ s.t. $B_i' \subseteq X_{a(i)} \quad \forall \alpha \in A$. define $\psi_\alpha = \sum_{i; a(i)=\alpha} g_i$

$\operatorname{Supp} \psi_\alpha = \overline{\bigcup_{i; a(i)=\alpha} B_i} = \bigcup_{i; a(i)=\alpha} \overline{B_i} \subseteq X_\alpha$. $\square$.

Rmk. Some ideas: 1) Why choose $B_i' \supseteq B_i$? Since we want to construct



~~should be all zero, it's "local"~~

$\varphi_i(\overline{B_i}) = \overline{B_{r_i}(0)}$ ~~to contract it's closure~~

$\subseteq B_{r_i'}(0)$ in the coordinate char

2) Why regular ? It's just came from the lemma where we construct such local smooth function on $B_{r_1}(0)$ & $B_{r_2}(0)$

3) Why local finitness ? Because we want $\forall x \in M$. $\sum \psi_\alpha(x)$ is a finite sum

4) In this lecture notes. $\operatorname{Supp}(f) := \overline{\{x \mid f(x) \neq 0\}}$.